

Material Database

nk_data folder

ALON.npy	ITO.npy	Ta2O5.npy
CdS.npy	Si.npy	TiO2.npy
MgF2.npy	SiO2.npy	...

Optical Properties Calculation

angle.py

Calculates
layer angles
from the
input angle
of incidence

cascaded_
matmul.py

Efficient
transfer matrix
multiplications
with
jax.lax.scan

data.py

Loads,
adds, and
interpolates
material data

fresnel.py

Calculates
reflection/
transmission
for s/p
polarized
light

plots.py

Visualizes
thin-film
structure
and fields

reflect_
transmit.py

Computes
reflectance and
transmittance
coefficients
of materials

tmm.py

Computes
thin-film
properties
using other
modules

wavevector.py

Computes
normal
wavevector and
incoherent
pass ratio

Analysis

analysis.py

Provides analysis tools for thin-film structures, including changes in transmittance, reflectance, absorbance, and color resulting from thickness variations